1 Non-dimensionalisation

in terms of $c = \sqrt{gH}$, gravitational wave speed and $Fr = \frac{U}{c}$, Froude number

$$\frac{\partial u}{\partial t} + c \cdot Fr \frac{\partial u}{\partial x} + c \frac{\partial h}{\partial x} = 0$$

$$\frac{\partial h}{\partial t} + c \cdot Fr \frac{\partial h}{\partial x} + c \frac{\partial u}{\partial x} = 0$$

which can be written as system

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ h \end{pmatrix} = -\underbrace{c \cdot Fr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{A_u}\begin{pmatrix} u \\ h \end{pmatrix} - \underbrace{c \cdot Fr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\frac{\partial}{\partial x}}_{A_h}\begin{pmatrix} u \\ h \end{pmatrix} - \underbrace{c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\frac{\partial}{\partial x}}_{F}\begin{pmatrix} u \\ h \end{pmatrix} - \underbrace{c\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\frac{\partial}{\partial x}}_{G}\begin{pmatrix} u \\ h \end{pmatrix}$$

2 Single wave-mode solution

$$\begin{pmatrix} u \\ h \end{pmatrix} = \begin{pmatrix} u_0 \\ h_0 \end{pmatrix} e^{i(kx - \omega t)}$$

$$\implies \frac{u_0}{h_0} = \pm 1$$

$$\omega = kc(Fr \pm 1)$$

for which the operators map

$$A_u \mapsto -ikcFr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A_h \mapsto -ikcFr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$F \mapsto -ikc \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$G \mapsto -ikc \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

3 Numerical dispersion

for a centred-difference scheme (where j denotes the spatial grid point)

$$\frac{\partial}{\partial x} y_j \rightarrow \frac{y_{j+1} - y_{j-1}}{2\Delta x}$$

for a wave mode, $y_j = e^{ikx_j}$for any j so

$$\begin{aligned}
\frac{\partial}{\partial x} y_j &\rightarrow \frac{1}{2\Delta x}(e^{ikx_j + \Delta x} - e^{ikx_j - \Delta x}) \\
&\rightarrow \frac{1}{2\Delta x}(e^{ik\Delta x} - e^{-ik\Delta x})e^{ikx_j} \\
&\rightarrow \frac{i}{\Delta x}\sin(k\Delta x)e^{ikx_j}
\end{aligned}$$

so, instead of the operators in section 2, the following should be used:

$$A_u \mapsto -\frac{i}{\Delta x}\sin(k\Delta x) c Fr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A_h \mapsto -\frac{i}{\Delta x}\sin(k\Delta x) c Fr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$F \mapsto -\frac{i}{\Delta x}\sin(k\Delta x) c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$G \mapsto -\frac{i}{\Delta x}\sin(k\Delta x) c \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

4 Formulating in terms of Courant numbers

Given the update for any iteration of a numerical scheme for the linear SWE can be written in the form $\begin{pmatrix} u^i \\ h^i \end{pmatrix} = M_1 \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + M_2 \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} = M \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$, the scheme is stable where the eigenvalues of matrix M, $|\lambda_\alpha| \leq 1$, so the problem of stability reduces to finding the region where this holds, in terms of the advective Courant number, c_a, and the gravitational Courant number, c_g.

$$\begin{aligned}
c_a &= \frac{c \cdot Fr\Delta t}{\Delta x} \\
c_g &= \frac{c\Delta t}{\Delta x} \\
\implies Fr &= \frac{c_a}{c_g} \text{ and} c = \frac{c_g \Delta x}{\Delta t}
\end{aligned}$$

Also let $\theta = k\Delta x$.
With these substitutions, the matrix M for any iteration of each numerical scheme can be written in terms of the variables c_a, c_g, and θ. Varying θ over $[0, 2\pi)$ yields all possible values of $\sin\theta$, and therefore the maximum of this, for any fixed c_a and $c_g$will indicate whether or not the scheme is stable for these values.